



Improved algorithms for extrinsic author verification

Nektaria Potha¹ · Efstathios Stamatatos¹ 

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Abstract

Author verification is a fundamental problem in authorship attribution, and it suits most relevant applications where it is not possible to predefine a closed set of suspects. So far, the most successful approaches attempt to sample the non-target class (all documents by all other authors) and transform author verification to a binary classification task. Moreover, they follow the instance-based paradigm (all documents of known authorship are treated separately). In this paper, we propose two algorithms, one instance-based and one profile-based (all known documents are treated cumulatively) that are able to outperform state-of-the-art methods in several benchmark datasets. We demonstrate that the proposed methods are capable of taking advantage of the availability of multiple documents of known authorship and that they are robust when text length is reduced.

Keywords Author verification · Authorship analysis · Stylometry · Text categorization · Text mining

1 Introduction

Nowadays, the categorization of texts in predefined classes has gained prominence due to the continuously increasing availability of digital documents in Internet media and the ensuing need of organizing and exploiting this information [2]. Topic, sentiment, and style are the main factors of distinguishing between texts. Style-based text categorization focuses on detecting the genre of documents or revealing the author(s) of documents. The latter is based on analyzing the personal writing style of authors. Stylometry (i.e., the quantification of style) is a far more challenging task in comparison to quantification of topic or sentiment since, usually, there are no specific words that can be associated with style. In addition, there is no consensus on what constitutes style, and there are plenty of stylometric approaches in the relevant literature [13,33].

✉ Efstathios Stamatatos
stamatatos@aegean.gr

Nektaria Potha
nekpota@aegean.gr

¹ Department of Information and Communication Systems Engineering, University of the Aegean, 83200 Karlovassi, Samos, Greece

In recent years, authorship analysis attracts constantly increasing attention and has been associated with multiple applications in areas such as digital text forensics (e.g., attribution of messages or proclamations to known terrorists, linking different messages by authorship, resolving copyright disputes, etc.) [1,38], humanities (e.g., attributing anonymous or disputed literary works to known authors) [14,37], and social media analytics (e.g., revealing multiple aliases of the same user in social media) [23,30].

Authorship attribution is the task of identifying the author of a questioned document, given a set of candidate authors and a number of document samples known to be written by each one of them. The vast majority of published studies focus on closed-set attribution, where a relatively small closed set of candidate authors is examined and the questioned documents should be written by one of them [33]. However, in multiple applications, it is not practical, or even possible, to define such a closed set of candidates. Imagine the case of a suspicious post in social media. Practically, anyone could be the author of such a text. Even when the set of candidates increases to include hundreds or thousands of authors, it is not at all certain that the true author will be ever included in this set. In such cases, a far more realistic solution is to adopt the open-set scenario, where the true author of the questioned document may not be included in the candidate set [20].

Authorship verification is a special case of open-set attribution, where the candidate set is singleton. It attempts to determine whether a specific author wrote a questioned document. Author verification is a fundamental task since all authorship attribution cases, both closed-set and open-set ones, can be decomposed into a set of authorship verification problems [22]. In addition, open-set attribution is more challenging when the candidate set is small [20], and therefore, author verification is a very challenging task.

In this paper, we adopt the definition of author verification as it was used in the recent PAN shared tasks [15,34,35]. In particular, given a set of documents of known authorship D_{known} , all written by the same author, the task is to estimate whether another document of unknown (or disputed) authorship d_{unknown} was also written by that author. Beyond a binary (yes/no) answer, an author verification system should also be able to provide a numerical score to indicate the certainty of the answer. In PAN shared tasks, this score was defined as the probability of a positive answer. Practically, a score of 0.5 corresponds to the case where the verification problem is left unanswered [35].

Author verification is a one-class classification task, where some samples of the target class are given (D_{known}) and the non-target class (all documents by all other authors) is extremely huge and heterogeneous. *Intrinsic verification methods* attempt to deal with this problem by analyzing only the given documents and estimating a similarity between d_{unknown} and D_{known} . When this similarity is high enough, then the unknown text is attributed to the author in question. On the other hand, *extrinsic verification methods* sample the non-target class and form a set of external documents D_{external} . Essentially, they attempt to transform author verification to a binary classification task and they estimate whether d_{unknown} is more similar to D_{known} rather than D_{external} . Figure 1 shows the difference between intrinsic and extrinsic author verification methods.

The most well-known and effective extrinsic verification approach is called *Impostors* and it was introduced by Koppel and Winter [22].¹ This method determines whether two documents are by the same author and builds a random subspace ensemble by randomly choosing a subset of features and a subset of external documents in each iteration. If the similarity between these two documents is found higher than their similarity with the external documents in the majority of repetitions, then a positive answer is provided. A generalized

¹ An early-view version of [22] was available since 2013.

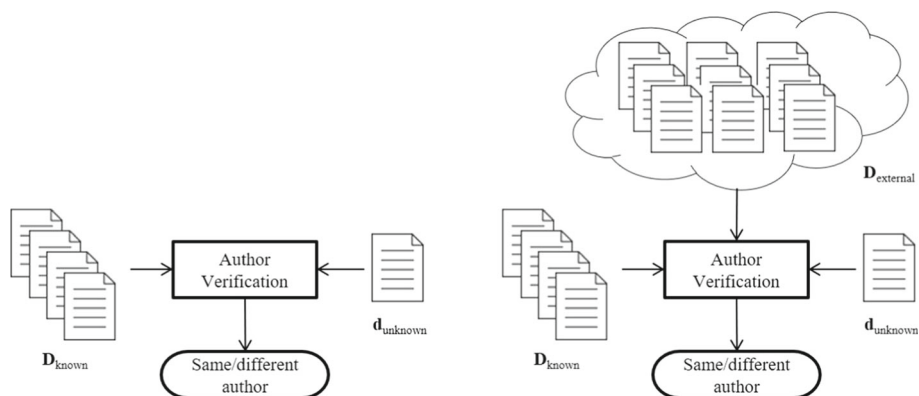


Fig. 1 Intrinsic (left) versus extrinsic (right) verification methods

version of this method, called *General Impostors* (GI) [32], that can also handle cases where $|D_{\text{known}}| > 1$, won the top-ranked position (among 18 participants) in PAN-2013 shared task in author verification [15]. In addition, a modification of GI [17] won the top-ranked position (among 13 participants) in the corresponding PAN-2014 shared task [35]. Finally, the second-best submission in PAN-2015 shared task (among 18 participants) was partially based on GI [25].

From another perspective, author verification methods can be distinguished as the ones following the *instance-based paradigm* and those following the *profile-based paradigm* [33]. The former is a document-centric approach and treats each document in D_{known} separately. On the other hand, the profile-based approach is author-centric and combines all documents in D_{known} , usually by concatenating them. The majority of the existing author verification methods, including GI, follow the instance-based paradigm [34,35].

This paper proposes extrinsic verification methods, inspired by GI, that notably enhance state-of-the-art performance. In particular, we study one instance-based version that enriches the information that is kept in each iteration when building the random subspace ensemble. This algorithm was introduced in [29]. In the current paper we present a more extensive study of its properties and performance. In addition, we introduce a profile-based version of GI that is able to take advantage of the case when multiple known documents are available. Based on an extensive experimental study using benchmark datasets that cover several languages, genres, and degrees of difficulty, we demonstrate that the latter is more effective in most of the cases. Furthermore, we demonstrate the effect of known document set size and text length in the performance of the proposed models.

The rest of this paper is organized as follows. In Sect. 2 an extensive review of previous author verification studies is presented, while Sect. 3 describes the proposed methods highlighting their differences with the original GI. Section 4 includes the experimental evaluation, and Sect. 5 discusses the main conclusions drawn from this study and possible future work directions.

2 Previous work

Until recently, authorship verification was a particularly under-explored task in authorship analysis. An early study by Stamatatos et al. [36] examined author verification along

with closed-set attribution and was based on NLP-based features and a multiple regression response function. A seminal study by Koppel et al. [21] introduced the *unmasking* method. It is based on a SVM classifier that distinguishes between samples taken from two documents, and it measures the drop in classification accuracy when the most important features are gradually removed. If the accuracy drops fast, then the two documents are by the same author. Unmasking is a very reliable author verification method when long texts (e.g., novels) are considered and it is not affected by topic (dis)similarities. However, it is unable to handle relatively short documents (e.g., articles) [31]. Furthermore, it is heavily confused in cross-genre conditions (e.g., when one document is literary prose and the other is a theater play [16]). Other early approaches are discussed in [7,24].

Research in author verification was strongly motivated by the recent PAN shared tasks that developed multiple benchmark datasets and established an evaluation framework for this task [15,34,35]. Multiple verification models have been submitted and tested in these evaluation campaigns. In general, authorship verification models can be distinguished into two main categories: intrinsic and extrinsic ones. As shown in Fig. 1, intrinsic models only use $(D_{\text{known}}, d_{\text{unknown}})$, and they attempt to handle a one-class classification task [10,12,28]. Usually, intrinsic models are quite fast and robust since they have to analyze a few documents and they do not depend on extrinsic factors. On the other hand, extrinsic models attempt to transform the verification task to a binary classification problem by sampling the negative class (D_{external}) [3,22,32]. Such models are usually more effective. A typical example is the *Impostors* method [22] and its modifications [17,32] that won top-ranked positions in PAN-2013 and PAN-2014 shared tasks. However, they are affected by the selection of the external documents. Koppel and Winter [22] demonstrate that when the external documents are in the same genre with the known/unknown documents, then the accuracy results are considerably improved.

With respect to the way the members of D_{known} are treated, verification models can be divided into instance-based and profile-based ones [33]. The former handle each known document separately and compare it with the unknown document [8,11,12,32]. This approach is document-centric and can exploit striking similarities of a certain known document with the unknown document even when other known documents are relatively distant. On the other hand, the profile-based paradigm is author-centric and combines (i.e., concatenates) all known documents into a long one and compares it with the unknown document [18,26,28]. This approach disregards differences between known documents, but it is more robust in case of short texts since it provides a more accurate representation of the concatenated known documents.

From another point of view, some methods require a training corpus (i.e., a number of author verification instances) and view author verification task as a binary classification task. Actually, they apply machine learning algorithms to learn to distinguish between positive (same author) and negative (different author) verification instances [4,8,11,26]. Naturally, the effectiveness of these models heavily depend on the representativeness and size of the training corpus. The evaluation results of PAN shared tasks demonstrated that such methods can be very effective when an appropriate training corpus is provided [34,35].

It is worth noticing that combining several verification models in a heterogeneous ensemble seems to be a very effective solution. The PAN organizers report improved evaluation results when combining all submissions [15,35]. Moreover, Moreau et al. [25] combine five verification approaches, including GI, and get very competitive results [34].

Most state-of-the-art verification methods are based on simple stylometric approaches that utilize character or lexical information. Popular features include character n -grams and frequent word frequencies [12,17,19,28]. The winner participant of PAN-2015 shared task

introduces a character-level recurrent neural network model [3]. More sophisticated models that involve syntactic analysis of texts are not competitive probably due to noise introduced by NLP tools used to provide syntactic parsing [34,35].

3 The proposed method

3.1 The general impostors method

The original GI method is essentially a random subspace ensemble [32] and it is a simple modification of the method introduced by Koppel and Winter [22] in order to handle multiple texts of known authorship. Initially, each document in D_{known} is examined separately. Within an iterative process, a subset of features and a subset of available impostors are selected in each repetition. The stylistic similarity of selected impostors with both the known and unknown documents is calculated and the number of times the known document was found more similar than all the impostors is recorded. Then, a verification score is calculated based on an aggregate function that combines all scores corresponding to the known documents. This is illustrated in Algorithm 1.

The similarity function used to estimate the stylistic closeness of two documents and the aggregate function that combines the scores for all known documents can be selected among several alternatives to optimize performance in a given training dataset [32].

3.2 Ranking-based impostors

The first proposed variation of GI has two main novelties:

1. Instead of selecting the impostors per problem randomly, the most relevant impostors to the known document are considered. That way, we increase the likelihood of using more appropriate impostors. To use a metaphor, in a police line-up, it does not make sense to use impostors with different characteristics, they all should be of roughly the same gender, age, height, race, etc., with the main suspect. Similarly, we consider the selected impostors as direct competitors of the known document and, therefore, it is now more challenging to decide whether the impostors or the known document are more similar to the unknown document. As a consequence, we only compare the impostors with the unknown document.
2. We exploit more information in each repetition. The original GI counts the number of times the known document is surpassed by at least one impostor. Thus, it disregards the number of impostors that surpassed the known document. However, this information is crucial. Imagine the case the known document is surpassed by exactly one (not always the same) impostor document in all repetitions. Then it would get a score of 0. In the proposed variation, we consider the rank of the known document within the set of impostors with respect to their similarity with the unknown document. The higher the ranking of the known document, the more likely we consider it to be written by the author of the unknown document. In the aforementioned example, the proposed method would provide a score of 0.5.

This method is called ranking-based impostors (RBIs), and it is illustrated in Algorithm 2. Note that it accepts exactly the same data and parameters in comparison with the original GI. In addition, to make sure that the length of impostor documents does not affect the

similarity calculation, all selected impostors are truncated according to the length of the known documents. In more detail, each selected impostor document is truncated separately for each known document. For a fair experimental comparison, this is applied to both our implementation of GI and RBI. Note also that both GI and RBI are stochastic algorithms (since they make some random choices), but RBI makes less stochastic choices since the most similar external documents will be considered as impostors for a given known document.

3.3 Profile-based impostors

The original GI method as well as RBI follow the *instance-based paradigm* [33], that is, they treat all documents in D_{known} separately. We propose another variation of this method that follows the *profile-based paradigm* where all known documents are concatenated and the differences between them are disregarded. That way, there is only one long known document (from which the *profile* of the author can be extracted) and there is no need to aggregate results for separate known documents. On the other hand, the resulted long known text should be compared with appropriate impostors. Given that this long text is composed of several constituent documents, a fact affecting its thematic homogeneity, the impostors should have similar characteristics.

The proposed profile-based impostors (PBI) method is demonstrated in Algorithm 3. It accepts the same data and parameters as GI and RBI do. In more detail:

- Only the external documents with the highest cumulative similarity to the set of known documents are considered ($\text{Impostors}_{\text{problem}}$). That way, external documents of distinct style and topic with respect to the known documents are filtered out. In contrast to RBI, the selected external documents should be similar enough to all known documents. That is, if an external document is very similar to just one known document but highly dissimilar to the rest of them, it is not likely to be selected since its cumulative similarity value will be relatively low. That way, the selected documents are more likely to have stylistic rather than thematic similarities with the known documents.
- Each impostor is constructed by selecting one seed document $\in \text{Impostors}_{\text{problem}}$ randomly. Then, its nearest neighbors in the set of impostors ($\text{Impostors}_{\text{problem}}$) are found so that we have exactly $|D_{\text{known}}|$ documents. The concatenation of these documents corresponds to an artificial impostor document that shares similar characteristics with the known text, namely it is composed of the same number of constituent documents, which have certain stylistic similarities.
- As in RBI, the ranking information of the known document within the set of impostors is considered in order to calculate the verification score.

As in GI and RBI, each impostor document is truncated based on the length of the known document (concatenation of all known documents) so that their size does not affect the calculation of similarity score.

4 Experimental study

4.1 Description of data

In the framework of PAN evaluation campaigns in authorship verification between 2013 and 2015, several corpora were built to cover various natural languages, genres and degrees

Data: $D_{known}, d_{unknown}, D_{external}$
Parameters: $repetitions, |Impostors_{problem}|, |Impostors_{repetition}|, rate$
Result: *VerificationScore*
for each $d_{known} \in D_{known}$ **do**
 Select $Impostors_{problem} \subset D_{external}$ randomly;
 Set $Score(d_{known}) = 0$;
 repeat $repetitions$ times
 Select $Impostors_{repetition} \subset Impostors_{problem}$ randomly;
 Select $rate\%$ of features randomly;
 for each $impostor \in Impostors_{repetition}$ **do**
 $Sim(impostor) = similarity(impostor, d_{known}) * similarity(impostor, d_{unknown})$
 end
 $Sim_{known} = similarity(d_{known}, d_{unknown})^2$
 end;
 if $Sim_{known} > \max(Sim(:))$ **then**
 $Score(d_{known}) = Score(d_{known}) + 1/repetitions$
 end
end
VerificationScore = *aggregate*(*Score*(:));

Algorithm 1: The original GI method [32]

Data: $D_{known}, d_{unknown}, D_{external}$
Parameters: $repetitions, |Impostors_{problem}|, |Impostors_{repetition}|, rate$
Result: *VerificationScore*
for each $d_{known} \in D_{known}$ **do**
 for each $impostor \in D_{external}$ **do**
 $relevance(impostor) = similarity(impostor, d_{known})$;
 end
 Select $Impostors_{problem} \subset D_{external}$ with highest $relevance(:)$;
 Set $Score(d_{known}) = 0$;
 repeat $repetitions$ times
 Select $Impostors_{repetition} \subset Impostors_{problem}$ randomly;
 Select $rate\%$ of features randomly;
 for each $impostor \in Impostors_{repetition}$ **do**
 $Sim(impostor) = similarity(impostor, d_{unknown})$;
 end
 $Sim_{known} = similarity(d_{known}, d_{unknown})$;
 Rank $S = Sim(:) \cup Sim_{known}$ in decreasing order;
 pos = position of Sim_{known} in S ;
 $Score(d_{known}) = Score(d_{known}) + 1/(repetitions * pos)$;
 end;
end
VerificationScore = *aggregate*(*Score*(:));

Algorithm 2: The proposed Ranking-Based Impostors method

of difficulty [15,34,35]. These corpora are considered as benchmark datasets for evaluating new authorship verification approaches. In this study, we use all datasets released in PAN-2014² and PAN-2015³ shared tasks (the corresponding datasets from PAN-2013 are of very small size). A PAN dataset consists of a collection of verification problems, each problem comprising a small set of known documents, all by the same author, and exactly one unknown document. Moreover, each problem is labeled as either positive (same author) or negative (different author). There are separate training and evaluation parts. All datasets are balanced

² <https://pan.webis.de/clef14/pan14-web/author-identification.html>.

³ <https://pan.webis.de/clef15/pan15-web/author-identification.html>.

Data: $D_{known}, d_{unknown}, D_{external}$
Parameters: $repetitions, |Impostors_{problem}|, |Impostors_{repetition}|, rate$
Result: $VerificationScore$
Set $relevance(:) = 0$;
for each $d_{known} \in D_{known}$ **do**
 for each $impostor \in D_{external}$ **do**
 $relevance(impostor) = relevance(impostor) + similarity(impostor, d_{known})$;
 end
end
Select $Impostors_{problem} \subset D_{external}$ with highest $relevance(:)$;
Set $VerificationScore = 0$;
 $knownText = concatenate(D_{known})$;
repeat $repetitions$ times
 Select $rate\%$ of features randomly;
 repeat $|Impostors_{repetition}|$ times
 Select $impostorSeed \in Impostors_{problem}$ randomly;
 Select $Neighbors \subset Impostors_{problem}$, the $|D_{known}| - 1$ most similar to $impostorSeed$;
 $newImpostor = concatenate(impostorSeed \cup Neighbors)$;
 $Sim(newImpostor) = similarity(newImpostor, d_{unknown})$;
 end;
 $Sim_{known} = similarity(knownText, d_{unknown})$;
 Rank $S = Sim(:) \cup Sim_{known}$ in decreasing order;
 $pos = \text{position of } Sim_{known} \text{ in } S$;
 $VerificationScore = VerificationScore + 1/(repetitions * pos)$;
end;

Algorithm 3: The proposed Profile-Based Impostors method

regarding the number of positive and negative instances, in both training and evaluation parts. All documents within a problem are in the same language and text length may vary. In some datasets, genre and thematic area is controlled (same for all documents within a problem) [35]. In other more challenging datasets, known and unknown documents are about distant topics or belong to different genres (cross-topic or cross-genre conditions) [34].

Table 1 provides a description of properties for each evaluation dataset (the corresponding training datasets have similar characteristics). The dataset properties are not homogeneous. For example, the number of known documents may be minimum (e.g., PAN14-EN), of varying size (e.g., PAN14-GR), or fixed (e.g., PAN14-SP). A document may consist of just a couple of paragraphs (e.g., PAN14-DR), a couple of pages (e.g., PAN14-GR), or several pages (e.g., PAN14-EN). All PAN14 datasets control both genre and thematic area, while PAN15 datasets are more challenging including cross-genre (e.g., PAN15-DU) and cross-topic (e.g., PAN15-GR) cases. More details on these datasets are provided in [34,35].

4.2 External documents

Given that the proposed methods need a set of external documents for each verification problem, we follow the practice of previous studies [17,32]. In particular, we form queries from the documents of the training corpus in each dataset, submit these queries in a search engine, and download impostor documents from the World Wide Web. In contrast to previous studies, our approach is only based on the known documents of the training corpus. Since the external documents are going to be used as competitors to the known documents, it makes sense to use information from the known documents to retrieve them. Moreover, in case we have a cross-topic or cross-genre dataset, then the properties of the known documents are quite different than the ones of the unknown documents.

Table 1 Statistics of PAN-2014 and PAN-2015 authorship verification evaluation datasets. $|D_{\text{known}}|$ and $|d|$ are average number of known documents and average text length (in words), respectively

Dataset	Language	Genre	Topic	#Problems	$ D_{\text{known}} $	$ d $
PAN14-DE	Dutch	Essays	Similar	96	2.0	398
PAN14-DR	Dutch	Reviews	Similar	100	1.0	116
PAN14-EE	English	Essays	Similar	200	2.6	833
PAN14-EN	English	Novels	Similar	200	1.0	6104
PAN14-GR	Greek	Articles	Similar	100	2.7	1537
PAN14-SP	Spanish	Articles	Similar	100	5.0	1121
PAN15-DU	Dutch	Mixed	Mixed	165	1.7	360
PAN15-EN	English	Plays	Mixed	500	1.0	536
PAN15-GR	Greek	Articles	Mixed	100	2.8	756
PAN15-SP	Spanish	Mixed	Mixed	100	4.0	946

In order to extract significant terms from the documents and form appropriate queries, we first perform clustering on the set of unique known documents D of the training corpus based on EM algorithm [5]. The resulting clusters C indicate basic thematic areas of those documents. Then, we extract the most significant terms from both documents and clusters. In more detail, the set of document-related terms T_{document} contains all terms t occurring in a document d when the following criteria are satisfied:

$$t \in T_{\text{document}} \Leftrightarrow \exists d \in D : f(t, d) \geq 2 \wedge df(t) \leq 3 \wedge |t| \geq 3 \quad (1)$$

where $f(t, d)$ is the raw frequency, $df(t)$ is the document frequency (the number of different documents the term occurs), and $|t|$ is the length (in characters) of term t (i.e., very short terms are excluded). Upon inspection, we could see that the selected terms were mostly thematic-related. In addition, the set of cluster-related terms T_{cluster} comprises all terms t occurring in a cluster c when the following criteria apply:

$$t \in T_{\text{cluster}} \Leftrightarrow \exists c \in C : f(t, c) \geq 2 \wedge cf(t) = 1 \wedge |t| \geq 3 \quad (2)$$

where $f(t, c)$ is the raw frequency of term t in cluster c and $cf(t)$ is the cluster frequency of t (the number of different clusters the term occurs). The extracted terms are characteristic of a certain cluster since they do not occur in other clusters. Then, several queries composed by exactly five terms are formed by randomly drawing (without replacement) from $T_{\text{document}} \cup T_{\text{cluster}}$. We use *Bing* search engine and download the first results of each query. These documents have mainly thematic similarities with the known documents. The downloaded documents are stripped from HTML tags and filtered out so that only relatively long documents (longer than the average text length of the respective dataset) remain. For each dataset, the upper limit of the external documents collected with the described method is 700. The number of external documents per dataset varies from 400 to 700 since it depends on the number of extracted significant terms (which affects the number of queries) and the results returned from the search engine.

4.3 Experimental setup

As can be seen in Algorithm 1, GI has several parameters that need to be tuned for a particular dataset. In previous work using PAN benchmark data, all these parameters were tuned based on the training corpus of each dataset separately [17,32]. In order to make this process more efficient, in this study we attempt to reduce independent parameters by associating $|\text{Impostors}_{\text{problem}}|$ with $|\text{Impostors}_{\text{repetition}}|$ and repetitions. Let $k = |\text{Impostors}_{\text{problem}}|$. Then, we set $|\text{Impostors}_{\text{repetition}}| = k/10$ and repetitions = $k/5$. Thus, by tuning k we set all these three parameters. For each dataset, we select $k \in \{50, 100, \dots, 400\}$ that optimizes performance in the training corpus. Moreover, we fix rate = 0.5 (i.e., 50% of the features are selected in each repetition). We use character 5-g as features based on some preliminary tests (a slight, though not consistent, difference with respect to the performance of character 3-g and 4-g was noticed). For each dataset, we first remove duplicate documents from the training dataset and sort character 5-g that occur in the remaining training documents by their frequency. Then, we use a frequency cutoff threshold to filter out the least frequent ones so that we get no more than 1500 features. That way, the frequency cutoff threshold is adapted to each dataset and we get a comparable number of features for all datasets. Finally, for each dataset, we select the *aggregation* function used in GI and RBI (among average, minimum, and maximum) and the *similarity* function (among cosine and minmax) that optimizes performance in the training corpus. These similarity functions are defined below:

$$\text{cosine}(\mathbf{x}, \mathbf{y}) = \frac{\sum_{i=1}^n x_i \cdot y_i}{\left(\sum_{i=1}^n x_i^2\right)^{\frac{1}{2}} - \left(\sum_{i=1}^n y_i^2\right)^{\frac{1}{2}}} \quad (3)$$

$$\text{minmax}(\mathbf{x}, \mathbf{y}) = \frac{\sum_{i=1}^n (\min(x_i, y_i))}{\sum_{i=1}^n (\max(x_i, y_i))} \quad (4)$$

Table 2 shows the parameter values selected for each PAN dataset. In most cases, minmax similarity and average aggregation provide the best results in the training corpus. This is in accordance to previous studies [32]. Note that the missing values in this table correspond to datasets with a single known document (see Table 1) where the aggregation function is not needed.

Since GI and the proposed methods make stochastic choices in each repetition, each experiment is repeated five times and we report average performance. Our main evaluation measure is the area under the ROC curve (AUC) that quantifies the effectiveness of an author verification approach for all possible verification score thresholds. Moreover, this evaluation measure does not depend on the distribution of positive/negative instances and was also used in PAN shared tasks [34,35]. In addition, we use c@1, a modification of classification accuracy, that is able to take into consideration cases left unanswered by a verification approach [27]. The main idea is to reward author verification methods that prefer to leave cases unanswered rather than providing wrong answers. This evaluation measure was also used in PAN shared tasks [34,35].

4.4 Results

In general, we follow exactly the same evaluation settings used in PAN shared tasks to achieve compatibility of comparison with PAN submissions [34,35]. In particular, all parameter values of GI, RBI, and PBI are estimated based on the training part of the dataset and the tuned

Table 2 The optimized parameter settings per dataset for the original GI method and the proposed variations (mmx, cos, and avg stand for minmax, cosine, and average respectively)

Dataset	GI			RBI			PBI	
	k	Sim.	Aggr.	k	Sim.	Aggr.	k	Sim.
PAN14-DE	100	mmx	avg	400	mmx	avg	350	cos
PAN14-DR	250	cos	–	150	mmx	–	100	mmx
PAN14-EE	150	mmx	min	150	mmx	min	100	mmx
PAN14-EN	250	mmx	–	400	cos	–	300	cos
PAN14-GR	150	mmx	max	300	mmx	max	350	cos
PAN14-SP	350	mmx	avg	350	mmx	avg	100	mmx
PAN15-DU	150	mmx	avg	50	mmx	avg	50	mmx
PAN15-EN	150	mmx	–	200	mmx	–	300	mmx
PAN15-GR	100	mmx	max	250	mmx	avg	300	mmx
PAN15-SP	200	mmx	avg	50	mmx	avg	150	mmx

models are evaluated using the test part of the dataset. Therefore, we can directly compare our results with those of the top-performing PAN-2014 and PAN-2015 participants. In more detail, we compare the performance of the proposed RBI and PBI methods with the following baseline methods:

- GI based on our implementation of Algorithm 1 and the same set of external documents used in RBI and PBI. This method was the top-performing submission in PAN-2013 shared task [32].
- Khonji and Iraqi (2014): This is a modification of GI that attained the overall top-ranked position in PAN-2014 shared task [17]. Similar to RBI, it attempts to enrich the information that is kept in each repetition. However, it takes into account the similarity scores that correspond to different known documents (while RBI considers their ranking position).
- Fréry et al. (2014): This method applies a decision tree learning algorithm in author verification by viewing the training corpus as a collection of positive and negative instances of a binary classification task [8]. It acquired the second top-ranked position in PAN-2014 shared task.
- Bagnall (2015): This is an extrinsic verification model based on a character-level multi-headed recurrent neural network model [3]. It was the top-ranked submission in PAN-2015 shared task.
- Moreau et al. (2015). This is an ensemble approach that is based on several heterogeneous verification models, including GI. It achieved very good results especially in the most challenging cross-genre PAN15-DU dataset and was ranked second overall in PAN-2015 shared task [25].
- Gutierrez et al. (2015): This is another modification of GI based on homotopy-based classification [9].

Tables 3 and 4 demonstrate the performance (AUC) of the proposed RBI and PBI methods together with the baseline models in PAN-2014 and PAN-2015 benchmark datasets, respectively. The performance results of all baseline methods except GI are taken directly from the PAN challenges [34,35]. As can be seen, in almost all cases both RBI and PBI are more effective than GI (only in PAN15-EN GI outperforms RBI). This shows that the proposed methods are clearly better options than the original GI. Moreover, in most of the cases, PBI

Table 3 AUC scores of the proposed and baseline models on PAN-2014 datasets

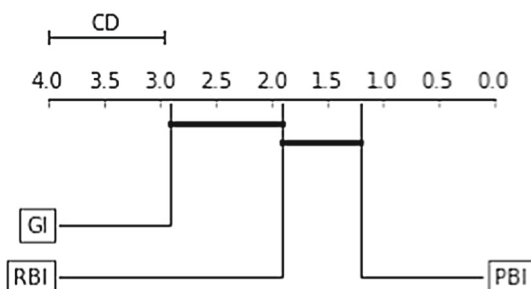
	PAN14-DE	PAN14-DR	PAN14-EE	PAN14-EN	PAN14-GR	PAN14-SP
Khonji and Iraqi (2014)	0.913	0.736	0.599	0.750	0.889	0.898
Fréry et al. (2014)	0.906	0.601	0.723	0.612	0.679	0.774
GI	0.947	0.660	0.618	0.730	0.772	0.606
RBI	0.975	0.759	0.762	0.767	0.929	0.878
PBI	0.951	0.791	0.754	0.790	0.969	0.903

Bold indicates the highest values per dataset (max value per column)

Table 4 AUC scores of the proposed and baseline models on PAN-2015 datasets

	PAN15-DU	PAN15-EN	PAN15-GR	PAN15-SP
Bagnall (2015)	0.700	0.811	0.882	0.886
Moreau et al. (2015)	0.825	0.709	0.887	0.853
Gutierrez et al. (2015)	0.592	0.739	0.802	0.755
GI	0.661	0.803	0.699	0.785
RBI	0.709	0.797	0.844	0.851
PBI	0.732	0.812	0.892	0.900

Bold indicates the highest values per dataset (max value per column)

Fig. 2 Results of a Nemenyi post-hoc test using both PAN-2014 and PAN-2015 datasets

outperforms RBI indicating that the profile-based paradigm better suits extrinsic verification. The comparison of the proposed methods with other state-of-the-art approaches reveals that in most of the cases (except PAN15-DU) they are able to improve the best results in each dataset.

In order to examine the statistical significance of differences in performance of the examined models in the whole collection of PAN datasets, we perform a nonparametric Friedman test and a Nemenyi post hoc test. Figure 2 presents the results of this test for GI, RBI, and PBI (the three models applied to all 10 PAN datasets) as it is described in [6]. For any pair of models the difference in their performance is statistically significant in case their scores differ more than the critical distance (CD) that was obtained for $p = 0.05$. As can be seen, PBI is statistically better than GI while there is no significant difference between PBI and RBI. Moreover, RBI is not significantly better than GI (their difference becomes significant at $p = 0.1$).

4.5 Contribution of ranking information

Next, we examine the contribution of one major novelty of the proposed methods, namely the ranking information used in each iteration. To be able to isolate the contribution of

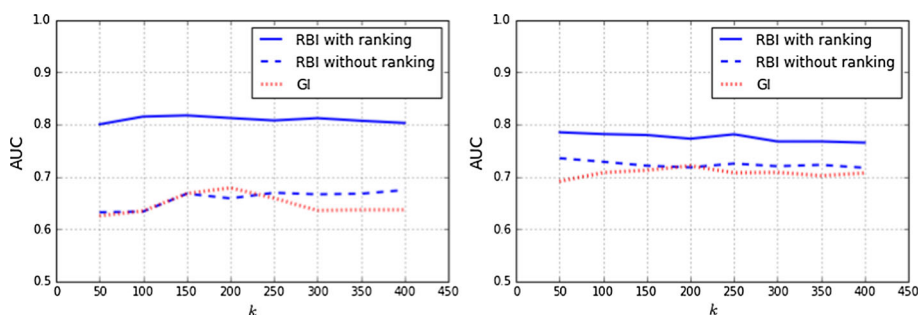


Fig. 3 Average performance (AUC) of RBI, with and without ranking information, as well as GI for PAN-2014 (left) and PAN-2015 (right) datasets

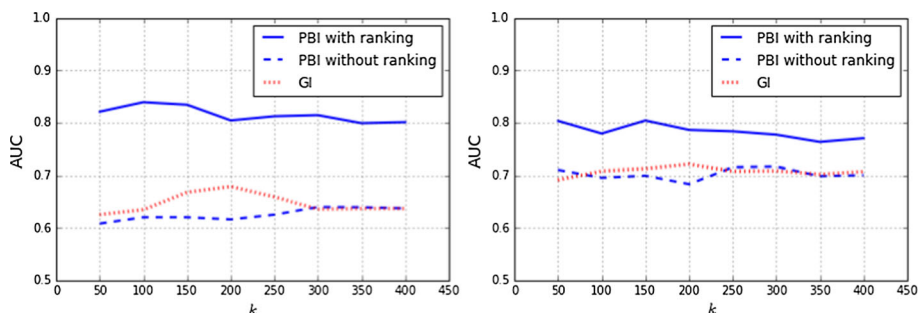


Fig. 4 Average performance (AUC) of PBI, with and without ranking information, as well as GI for PAN-2014 (left) and PAN-2015 (right) datasets

this factor, beyond Algorithms 2 and 3, we also use a modified version of them where the ranking information is not used (the decision rule of original GI is used to update *Score* and *VerificationScore* values).

We apply RBI and PBI with and without considering ranking information for varying values of parameter k (the similarity function is fixed to *minmax* and the aggregation function is fixed to *average* for this experiment). Each method is applied to each test dataset separately and then average AUC is shown for all PAN-2014 (Fig. 3) and all PAN-2015 datasets (Fig. 4), respectively. These figures also show the corresponding performance of GI for comparative purposes. It is apparent that the contribution of ranking information is significant and assists both RBI and PBI to enhance their effectiveness. The performance of both RBI and PBI without ranking is very similar to GI. Comparing the performance on PAN-2014 and PAN-2015 datasets, we see that the contribution of ranking is stronger for PAN-2014 while the version of the algorithms without ranking is weaker in PAN-2014. This indicates that when there are cross-topic and cross-genre conditions, ranking information is less crucial. Moreover, relatively low values of k (up to 150) achieve the best results in all experiments.

4.6 Size of known document set

In this section, we examine how the proposed methods are affected by the known document set size ($|D_{\text{known}}|$). In case multiple known documents are available, the verification method should take advantage of them and enhance its performance. Given that PAN datasets are not

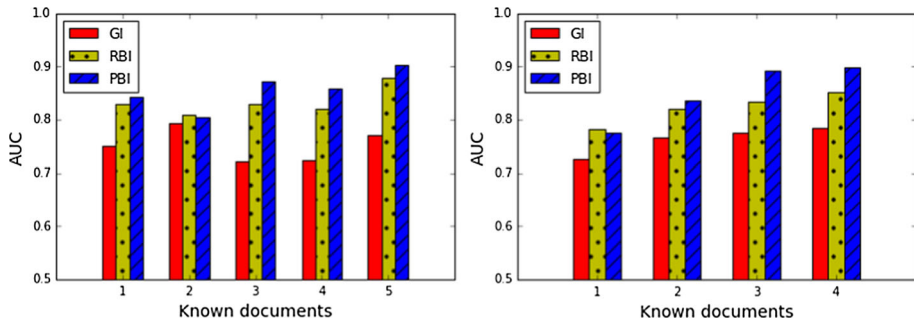


Fig. 5 AUC of GI, RBI, and PBI on PAN14-SP (left) and PAN15-SP (right) for varying number of known documents

homogeneous in terms of $|D_{\text{known}}|$ (see Table 1), we focus on two datasets that have a fixed size of known documents. In more detail, PAN14-SP and PAN15-SP have a fixed $|D_{\text{known}}| = 5$ and 4, respectively. By varying the size of known document set of these datasets from 1 to 5 (or 4) we are, then, able to study how this affects the examined algorithms.

Figure 5 depicts the performance of GI, RBI, and PBI on the test parts of PAN14-SP and PAN15-SP for varying $|D_{\text{known}}|$. The depicted results concern the best k values for each dataset estimated based on the corresponding training part. As previously, the similarity function is fixed to *minmax* and the aggregation function is fixed to *average*. With respect to GI, it achieves its best result on PAN14-SP for $|D_{\text{known}}| = 2$ while in the case of PAN15-SP it seems to get improved a little bit when $|D_{\text{known}}|$ increases. On the other hand, both RBI and PBI are generally improving when there are more known documents. RBI seems to be competitive in cases with a single or a couple of known documents, while PBI is superior when $|D_{\text{known}}| \geq 3$.

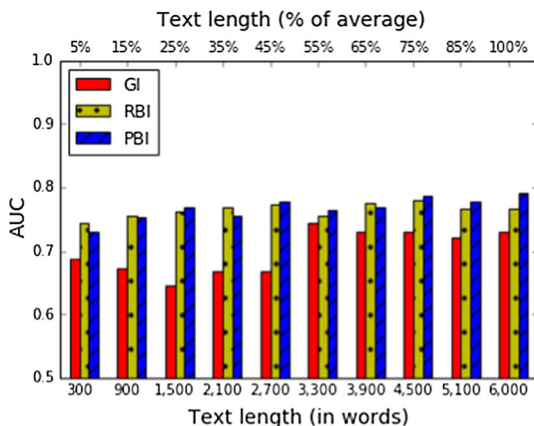
These results demonstrate that both RBI and PBI are better than GI when multiple known documents are available. Note that this is also indicated in the results presented in Tables 3 and 4 where the difference between GI and RBI/PBI is relatively low in datasets with low $|D_{\text{known}}|$ (e.g., PAN14-EN and PAN15-EN), while it becomes much larger in datasets with relatively high $|D_{\text{known}}|$ (e.g., PAN14-GR, PAN15-GR).

4.7 Effect of text length

An important factor that affects the performance of author verification (as well as authorship analysis) methods is text length of considered documents [33]. In order to study how text length affects the proposed methods, we focus on PAN14-EN, the dataset with the highest mean text length (see Table 1). Figure 6 shows the performance of GI, RBI, and PBI on the test part of PAN14-EN when text length of all documents within a verification problem decreases from 100 (ca. 6000 words) to 5% (ca. 300 words). Similar to the previous experiment, k values are optimized based on the corresponding training parts while *minmax* and *average* are used as similarity and aggregation functions.

In general, it seems that both RBI and PBI tend to improve with increasing text length. RBI achieves its maximum performance at ca. 2700 words (45% of the average length) while PBI performs best with maximal length (ca. 6000 words). It is also remarkable that RBI is slightly better than PBI for relatively short texts while PBI is better for long texts. The performance of GI is worse than that of both RBI and PBI. In addition, it is less robust since it is competitive with minimal text length (ca. 300 words), and then it seems to lose potential for longer texts.

Fig. 6 AUC of GI, RBI, and PBI on PAN14-EN for varying text length (showed both as percentage of average length and in words)



It almost reaches RBI and PBI for texts ca. 3300 words and then their difference becomes larger for longer texts.

4.8 Evaluation based on $c@1$

So far, all experiments used AUC as the evaluation measure. This is extracted by examining the ranking of verification scores (rather than their exact values) produced when a method is applied to a dataset. However, when one has to decide about a specific verification case, the verification score has to be transformed to a binary answer: either a positive (same author) or a negative (different author) one. To this direction, a threshold can be applied to the verification score (actually, the calculation of AUC is based on all possible thresholds). In order to set this threshold, information about the distribution of positive and negative verification instances is paramount. In all PAN datasets, positive and negative instances are equally distributed and this information was actually given to PAN participants.

In this paper, we examine three approaches to transform verification scores to binary answers. The first one is based on the verification scores obtained using the training part of each dataset. In particular, taking into account the balanced distribution of classes, the threshold is set to the median verification score of the training part of the dataset. Then, all verification scores of the test dataset that are lower/higher than this threshold are transformed to negative/positive answers. In the special case a verification score in the test dataset is exactly equal to the threshold value, that test problem is left unanswered. This is in accordance to the setup of PAN shared tasks and the used evaluation measure ($c@1$) is able to take such cases into account [34,35]. The second approach is very similar, but it is based on the median verification score of the test cases [3]. Intuitively, the estimation of the threshold based on the test cases should be more reliable (in comparison to that of the training problems). However, this approach requires a set of test cases and it is unable to handle a single verification case. The third method uses a logistic regression classifier trained on the verification scores of the training part of the dataset. This classifier is then applied to the test dataset, and it does not leave problems unanswered. The median threshold (MT) method, based on either the training or the test part of the dataset, and logistic regression (LR) can be combined with GI, RBI, or PBI to provide a complete author verification system.

The performance of the approaches proposed in this study as well as the baselines in terms of $c@1$ is shown in Tables 5 and 6. As can be seen, the RBI and PBI methods are in most of

Table 5 c@1 scores of the proposed and baseline models on PAN-2014 datasets

	PAN14-DE	PAN14-DR	PAN14-EE	PAN14-EN	PAN14-GR	PAN14-SP
Khonji and Iraqi (2014)	0.844	0.650	0.583	0.610	0.810	0.778
Frery et al. (2014)	0.906	0.578	0.710	0.588	0.642	0.750
GI-MT (training)	0.896	0.670	0.598	0.680	0.660	0.628
RBI-MT (training)	0.917	0.700	0.690	0.730	0.900	0.780
PBI-MT (training)	0.896	0.700	0.675	0.750	0.900	0.830
GI-MT (test)	0.914	0.610	0.595	0.693	0.744	0.613
RBI-MT (test)	0.938	0.712	0.650	0.745	0.921	0.799
PBI-MT (test)	0.940	0.710	0.680	0.730	0.923	0.837
GI-LR	0.896	0.637	0.575	0.684	0.747	0.595
RBI-LR	0.917	0.712	0.690	0.734	0.909	0.780
PBI-LR	0.900	0.713	0.675	0.724	0.911	0.830

Bold indicates the highest values per dataset (max value per column)

Table 6 c@1 scores of the proposed and baseline models on PAN-2015 datasets

	PAN15-DU	PAN15-EN	PAN15-GR	PAN15-SP
Bagnall (2015)	0.644	0.757	0.851	0.814
Moreau (2015)	0.770	0.638	0.781	0.775
Gutierrez et al. (2015)	0.556	0.694	0.725	0.674
GI-MT (training)	0.582	0.744	0.640	0.690
RBI-MT (training)	0.667	0.775	0.740	0.780
PBI-MT (training)	0.630	0.766	0.841	0.827
GI-MT (test)	0.624	0.781	0.653	0.764
RBI-MT (test)	0.640	0.771	0.817	0.797
PBI-MT (test)	0.648	0.785	0.863	0.834
GI-LR	0.637	0.782	0.661	0.778
RBI-LR	0.674	0.784	0.857	0.803
PBI-LR	0.678	0.776	0.868	0.840

Bold indicates the highest values per dataset (max value per column)

the cases more effective than the baselines. When either RBI or PBI is combined with MT or LR they constantly outperform the respective GI versions. In average, the MT(test) approach is more effective, as expected, since it handles all the test cases together. However, LR seems more effective in PAN-2015 datasets.

It has to be noted that it is possible to attempt to improve c@1 scores by leaving unanswered all cases with verification scores very close to the threshold [3]. In this paper, our aim is not to optimize performance in terms of c@1, and we focus on the relative improvement of the proposed methods RBI and PBI with respect to GI.

5 Conclusions

In this paper, we studied two improved algorithms for extrinsic author verification. RBI is a modification of GI that takes into account ranking information of the known document

among the impostors. In addition, PBI is a profile-based approach that concatenates all available known documents and constructs impostor documents with similar characteristics. The experimental results based on several PAN benchmark datasets show that the proposed methods are better than GI and other state-of-the-art methods based on both AUC (focusing on the ranking of verification scores) and c@1 evaluation measures (focusing on binary answers).

The contribution of the ranking information, used in both RBI and PBI, is a crucial factor to improve performance. This enables the proposed methods to take advantage when the known document is constantly among the top positions despite the fact that some impostor documents surpass it (probably based on irrelevant thematic similarities with the unknown document). Note also that ranking is a factor that can be easily used to combine information from different known documents. In a similar attempt, Khonji & Iraqi (2014) combine the similarity scores for each known document. However, the exact values of similarity scores depend on several factors, including text length, topic, etc., while ranking is more general.

The proposed methods are able to take advantage of the situation when multiple known documents are available. Their performance in datasets where $|D_{\text{known}}|$ is relatively high is notably better than GI. RBI seems to be better able to handle cases where $|D_{\text{known}}| < 3$ while PBI is more effective when $|D_{\text{known}}| \geq 3$. This indicates that the profile-based paradigm is a better option for large known document sets. Moreover, RBI is slightly better than PBI for relatively short texts, while PBI performs better with long texts.

All extrinsic verification methods strongly depend on the appropriate selection of D_{external} . In this paper, we used a simple approach that downloads external documents from the Web ensuring that there are thematic similarities with the known documents of the training corpus. Certainly, this procedure can be improved by taking into account the genre of documents. Ideally, the external documents should be sampled from the same source as the one from which the known documents are drawn. For instance, if the known documents are articles of a certain newspaper, then the external documents should also be articles from that newspaper to ensure similarity in genre, format, and edition of documents. However, information about the source of known documents may not be available.

The current work uses character n -gram features, a typical choice in similar studies [33]. An interesting future work direction could focus on a richer feature set, comprising character, lexical, and syntactic features, that can be adapted to each verification problem separately by considering the specific properties of known and unknown documents. This is especially crucial in challenging cross-genre conditions.

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Nektaria Potha is currently a postdoctoral fellow at the Department of Information and Communication Systems Engineering, University of the Aegean, Greece. She received her Ph.D. degree in computer science and a master's degree in Technologies and Management of Information and Communication Systems from the University of the Aegean and a degree in Physics from the University of Ioannina, Greece. Her main research interests are text mining, time series analysis, and machine learning focusing, in particular, on authorship verification and cyber-bullying detection.



Efsthios Stamatatos is currently an associate professor in the Department of Information and Communication Systems Engineering, University of the Aegean, Greece. He received a diploma in electrical engineering (1996) and a doctoral degree in electrical and computer engineering (2000), both from University of Patras. His main research interests include text mining and intelligent information processing with a particular focus on text classification, authorship analysis, and computational stylometry. He is a co-organizer of the PAN series of shared tasks in digital text forensics.